

Section: A

QUIZ - 2

Time Allowed: 20 mins

Name

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1. The cross-entropy loss used for binary logistic regression is also known as:
a) Hinge loss b) Mean absolute error ☒ c) Log loss (binary cross-entropy) d) Exponential loss
2. The gradient descent update for parameter θ (theta) can be written as:
a) $\theta := \theta + \alpha \cdot \nabla L$ ☒ b) $\theta := \theta - \alpha \cdot \nabla L$ c) $\theta := \alpha \cdot \nabla L$ d) $\theta := \nabla L \cdot \alpha$
3. A confusion matrix allows you to compute which of the following directly?
a) Learning rate ☒ b) Precision, recall, and F1 score c) Model weights d) Feature importance
4. In binary logistic regression, the hypothesis output (after sigmoid) is interpreted as:
a) Distance to the decision boundary
☒ b) A probability estimate of the positive class
c) The class label 0 or 1 directly
d) The residual error
5. Recall (sensitivity) measures:
a) The proportion of predicted positives that are actually positive
☒ b) The proportion of actual positives that were correctly predicted
c) The proportion of total samples correctly classified
d) The proportion of negative samples misclassified
6. A classifier that achieves high recall but low precision tends to:
☒ a) Miss many positive examples
b) Produce many false negatives
c) Predict the positive class too often
d) Have perfect balance between errors
7. A kNN classifier with $k=1$ is likely to: *single underfitting - high bias*
a) Have high bias b) Have high variance ☒ c) Always misclassify d) Underfit
8. When using kNN, feature scaling is important because:
☒ a) Distance metrics are sensitive to feature magnitude differences
b) The algorithm requires normally distributed data
c) It eliminates the need for selecting k
d) It prevents label imbalance
9. kNN is best described as a(n):
☒ a) Parametric algorithm
☒ b) Non-parametric, instance-based algorithm
c) Regularized linear classifier
d) Probabilistic generative model
10. A drawback of kNN is that it:
a) Requires training optimization
☒ b) Is slow for large datasets at prediction time
c) Cannot handle non-linear decision boundaries
d) Is insensitive to feature scaling

11. Information Gain for a candidate split is computed as:

- a) The entropy after split only
- ☒ b) The difference in entropy between parent and weighted children
- c) The Gini impurity multiplied by number of features
- d) The variance of target variable

12. The F1-score can be interpreted as:

- a) The arithmetic mean of precision and recall
- b) The harmonic mean of precision and recall
- ☒ c) The ratio of recall to precision
- d) The geometric mean of precision and recall

13. A perfectly pure leaf node has which of the following properties?

- a) It contains examples from multiple classes equally
- ☒ b) It contains only examples from a single class
- c) It is always the root of the tree
- d) It has maximum entropy

14. Which is TRUE for a very deep decision tree trained without constraints?

- a) Low variance, high bias
- ☒ b) High variance, low bias
- c) Low bias, low variance
- d) High bias, moderate variance

Overfitting - high variance, low bias.
Underfitting - high bias, low variance.

15. Decision trees are prone to overfitting especially when:

- a) They are shallow and under-parameterized
- ☒ b) Trees are grown to full depth on noise-prone datasets
- c) Data is plentiful and clean
- d) Features are continuous only

16. Random forests use feature bagging (random subset of features per split) primarily to:

- a) Increase correlation among trees
- ☒ b) Decorrelate trees and improve ensemble diversity
- c) Reduce number of classes
- d) Enforce linear splits

17. A model that performs poorly on both training and test data likely has:

- a) High variance
- ☒ b) High bias
- c) Low bias
- d) Perfect generalization

18. A Receiver Operating Characteristic (ROC) curve plots:

- a) Precision vs. recall
- ☒ b) Recall (TPR) vs. false positive rate (FPR)
- c) Accuracy vs. precision
- d) True negatives vs. true positives

19. A random forest's final binary prediction is typically produced by:

- ☒ a) Majority voting across trees
- b) Selecting the prediction of the first tree only
- c) Averaging leaf depths across trees
- d) Taking the maximum tree depth as threshold

20. Ensemble methods are especially useful when:

- a) Base learners are identical and make the same errors
- ☒ b) Base learners are weak but diverse, so errors cancel out
- c) A single model already achieves perfect generalization
- d) Data is entirely noise-free